

→ for discrete functions.

Z-Transformation:

Suppose $f[n]$ is a discrete function of variable n .

Then two-sided z-transformation is defined as

$$Z(f[n]) = \sum_{n=-\infty}^{\infty} f[n] z^{-n} ; \text{ where } n \text{ indicates an integer \& } z \text{ is a}$$

$$Z(f[n]) = \sum_{n=0}^{\infty} f[n] z^{-n} \quad \text{complex number i.e. } z = x + iy.$$

↳ only one sided.

$$= F(z)$$

* Z-transformation is the discrete version of Laplace transformation.

Find the z-transformation for the following function.

Example #1

$$u[n] = \begin{cases} 1 & ; n \geq 0 \\ 0 & ; \text{e.w.} \end{cases}$$

$n = 0, 1, 2, 3, 4, \dots$

$$Z(u[n]) = \sum_{n=0}^{\infty} (1) z^{-n} = 1 + z^{-1} + z^{-2} + z^{-3} + \dots$$

$$\frac{1}{1-(z^{-1})} ; |z^{-1}| < 1 \therefore S_z \frac{a}{1-r} ; |r| < 1$$

$$Z(u[n]) = \frac{1}{1-\frac{1}{z}} = \frac{z}{z-1} ; |z^{-1}| < 1 \text{ or } |z| > 1$$

∞ ∞

kernel transformation means
a function that converts discrete
to continuous.

Example #2

$$f[n] = \begin{cases} a^n & ; n \geq 0 \\ 0 & ; \text{e.w} \end{cases}$$

$$Z(f[n]) = \sum_{n=0}^{\infty} f[n] z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} \frac{a^n z^{-n}}{a^n} = \sum_{n=0}^{\infty} \left(\frac{z}{a}\right)^{-n}$$

$$= \left(\frac{z}{a}\right)^0 + \left(\frac{z}{a}\right)^{-1} + \left(\frac{z}{a}\right)^{-2} + \dots$$

$$\frac{a}{1-r} = \frac{1}{1-\left(\frac{z}{a}\right)^{-1}} ; \left|\left(\frac{z}{a}\right)^{-1}\right| < 1$$

$$\Rightarrow |z| > |a|$$

$$Z(f[n]) = \frac{1}{1 - a/z} = \frac{z}{z - a}$$

Example #3

$$f[n] = \begin{cases} a & ; n \geq 0 \\ 0 & ; \text{e.w} \end{cases}$$

$$Z(f[n]) = \sum_{n=0}^{\infty} f[n] z^{-n} = \sum_{n=0}^{\infty} a z^{-n}$$

$$= a \sum_{n=0}^{\infty} z^{-n}$$

$$= a \left[\frac{z}{z-1} \right]$$

Example # 4

$$f[n] = e^{an}; n \geq 0$$

$$Z(f[n]) = \sum_{n=0}^{\infty} f[n] z^{-n}$$

$$\sum_{n=0}^{\infty} (e^a)^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (e^{-a} z)^{-n}$$

$$= 1 + (e^{-a} z)^{-1} + (e^{-a} z)^{-2} + \dots$$

$$= \frac{1}{1 - (e^{-a} z)^{-1}}; |(e^{-a} z)^{-1}| < 1$$

$$= \frac{1}{1 - \frac{e^a}{z}}; \left| \frac{1}{e^{-a} z} \right| < 1$$

$$Z(f[n]) = \frac{z}{z - e^a} \Rightarrow |z| > e^a$$

Example:

$$f[n] = \frac{u[n]}{n!}$$

$$Z(f[n]) = Z\left(\frac{u[n]}{n!}\right)$$

$$= \sum_{n=0}^{\infty} \frac{u[n]}{n!} z^{-n}$$

$$= \frac{1}{0!} z^{-0} + \frac{1}{1!} z^{-1} + \frac{1}{2!} z^{-2} + \dots$$

$$\mathcal{Z}\left(\frac{u[n]}{n!}\right) = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!}$$

$$f(x) = e^{-x}$$

$= e^{-2}$; From Maclaurin Series $f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$

Example # 6

$$f[n] = u[n] \cos bn \quad ; n \geq 0$$

$$\mathcal{Z}(f[n]) = \mathcal{Z}(u[n] \cos bn)$$

$$\sum_{n=0}^{\infty} u[n] \cos bn z^{-n}$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\sum_{n=0}^{\infty} u[n] \left[\frac{e^{ibn} + e^{-ibn}}{2} \right] z^{-n}$$

$$= \frac{1}{2} \left\{ \sum_{n=0}^{\infty} u[n] e^{ibn} z^{-n} + \sum_{n=0}^{\infty} u[n] e^{-ibn} z^{-n} \right\}$$

from example 4 — do not use in exam, solve it with all steps.

$$= \frac{1}{2} \left[\frac{z}{z - e^{ib}} + \frac{z}{z - e^{-ib}} \right]$$

$$\mathcal{Z}(f[n]) = \frac{z^2 - z \cos b}{z^2 - z \cos b + 1}$$

using the formula

$$\cos b = \frac{e^{ib} + e^{-ib}}{2}$$

dobry sinbn.

$$f[n] = \delta(n - n_0)$$

sd:

$$Z(f[n]) = Z(\delta(n - n_0))$$

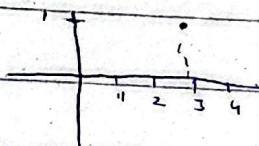
$$= \sum_{n=0}^{\infty} \delta(n - n_0) z^{-n} = 0 + 0 + 0 + \dots (1) z^{-n_0} + 0 + 0 + \dots$$

$$\S Z(f[n]) = z^{-n_0}$$

aka impulse function.

$$\delta(n - n_0) = \begin{cases} 1, n = n_0 \\ 0, \text{ew} \end{cases}$$

Example $Z(\delta(n - 3)) = z^{-3}$; $z = x + iy$



$$\begin{aligned} f[n] &= na^{n+3} u(n) + \delta(n) & Z(n) &= Z(n-0) \\ Z(f[n]) &= Z(\S na^{n+3} u(n)) + Z(\delta(n)) & &= \begin{cases} 1 & 0 \\ 0 & \text{ew} \end{cases} \\ & & + z^{-0} &= 1 \\ &= \sum_{n=0}^{\infty} na^{n+3} u(n) z^{-n} + 1 \end{aligned}$$

$$Z(n) = \frac{z}{(z-1)^2}$$

the something

$$f[n] = n \quad \text{or} \quad f[n] = u[n]n$$

$$Z(f[n]) = \sum_{n=0}^{\infty} f[n] z^{-n} = \sum_{n=0}^{\infty} n z^{-n}$$

$$Z(1) = \sum_{n=0}^{\infty} (1) z^{-n} = \frac{z}{z-1} \quad ; \text{ check } Z\text{-transformation of a constant.}$$

Diff wrt z

$$\frac{d}{dz} \sum_{n=0}^{\infty} z^{-n} = \frac{d}{dz} \left(\frac{z}{z-1} \right)$$

$$\Rightarrow \sum_{n=0}^{\infty} (-n) z^{n-1} = \frac{(z-1) \frac{d}{dz}(z) + z \frac{d}{dz}(z-1)}{(z-1)^2}$$

$$\begin{aligned} \neq \sum_{n=0}^{\infty} n z^{-n} z^{-1} &= \frac{(z-1)(1) - z(1)}{(z-1)^2} \\ &= \neq \frac{1}{(z-1)^2} \end{aligned}$$

$$z^{-1} \sum_{n=0}^{\infty} n z^{-n} = \frac{1}{(z-1)^2}$$

$$\sum_{n=0}^{\infty} n z^{-n} = \frac{z}{(z-1)^2}$$

$$Z(n) = \frac{z}{(z-1)^2}$$

not more than n^1 .

Example

$$f(n^2) = ?$$

$$\sum_{n=0}^{\infty} n 2^{-n} = \frac{2}{(2-1)^2}$$

$$f(n^2) =$$

$$\delta(n-n_0) = \begin{cases} 1 & ; n \geq n_0 \\ 0 & ; n \neq n_0 \end{cases}$$

$$Z(f[n]) = \sum_{n=0}^{\infty} f[n] z^{-n}$$

~~$$Z(f[n])$$~~

$$Z(\delta(n-n_0)) = \sum_{n=0}^{\infty} \delta(n-n_0) z^{-n}$$

$$= (0)z^{-0} + (0)z^{-1} + (0)z^{-2} + \dots + (1)z^{-n_0} + (0)z^{-(n_0+1)} + \dots$$

$$Z(\delta(n-n_0)) = z^{-n_0}$$

$$Z(\delta(n)) = 1 \quad \delta(n) = \begin{cases} 1 & ; n = 0 \\ 0 & ; n \neq 0 \end{cases} \quad \delta(n) = \begin{cases} 1 & ; n \geq 0 \\ 0 & ; n < 0 \end{cases}$$

Example

$$f(n) = n a^{n+1} u(n) + 4 \delta(n)$$

$$= \sum_{n=0}^{\infty} \{ n a^{n+1} u(n) + 4 \delta(n) \} z^{-n}$$

$$= \sum_{n=0}^{\infty} \{ n a^{n+1} z^{-n} \} + \sum_{n=0}^{\infty} \{ 4 \delta(n) z^{-n} \}$$

$$= a^2 \sum_{n=0}^{\infty} n a^n z^{-n} + 4 \sum_{n=0}^{\infty} \delta(n) z^{-n} \rightarrow i$$

$$= a^2 \sum_{n=0}^{\infty} n \left(\frac{z}{a} \right)^n + 4(1)$$

$$= a^2 \left(\frac{z/a}{(z/a-1)^2} \right) + 4$$

for 1-sided
Z transformation
it will not effect.

$$+ 4 \left(\frac{\frac{2}{9}}{\frac{(z-9)^2}{9^2}} \right) + 4$$

$$\frac{9^2 z}{(z-9)^2} + 4$$

Find the inverse z transformation of the following function.

$$F(z) = \frac{z}{z^2 + 7z + 10}$$

$$z(f[n]) = \frac{z}{z^2 + 7z + 10}$$

~~$$z(f[n]) =$$~~

$$f[n] = z^{-1} \left\{ \frac{z}{(z+5)(z+2)} \right\}$$

→ use partial fraction.

$$= z^{-1} \left\{ z \cdot \frac{1}{(z+5)(z+2)} \right\}$$

$$= z^{-1} \left\{ z \left(\frac{1}{3(z+2)} - \frac{1}{3(z+5)} \right) \right\}$$

$$= \frac{1}{3} \left\{ z^{-1} \left(\frac{z}{z-(-2)} \right) - z^{-1} \left(\frac{z}{z-(-5)} \right) \right\}$$

$$= \frac{1}{3} \left\{ (-2)^n - (-5)^n \right\}$$

Difference Equation

z-transformation of difference operator.

$$z(f[n+1]) = \sum_{n=0}^{\infty} f[n+1] z^{-n}$$

$$f[1]z^{-0} + f[2]z^{-1} + f[3]z^{-2} + \dots$$

$$\text{let } m = n+1 \Rightarrow n = m-1$$

$$\text{When } n=0, \text{ then } m=0+1=1$$

$$\text{When } n=\infty, \text{ then } m=\infty+1=\infty$$

$$\text{Eqn 1} \Rightarrow z(f[n+1]) = \sum_{m=1}^{\infty} f[m] z^{-(m-1)}$$

$$\Rightarrow z(f[n+1]) = z \sum_{m=1}^{\infty} f[m] z^{-m}$$

$$= z \left[f[1]z^{-1} + f[2]z^{-2} + f[3]z^{-3} + \dots + \underbrace{f[0]z^{-0} + f[0]z^{-1}}_{\text{Addition \& Subtraction}} \right]$$

$$z(f[n+1]) = z \left[f[0]z^{-0} + f[1]z^{-1} + f[2]z^{-2} + \dots - f[0]z^{-0} \right]$$

$$= z \left[\sum_{n=0}^{\infty} f[n] z^{-n} - f[0] z^{-0} \right]$$

$$= z \left[z(f[n]) - f[0](1) \right]$$

$$z(f[n+1]) = z z(f[n]) - z f[0]$$

$$\mathcal{Z}(f[n+2]) = z^2 \mathcal{Z}(f[n])$$

$$= z^2 f[n+1] + z^2 f[0] - 2f[1]$$

$$\mathcal{Z}(f[n-1]) = \sum_{n=0}^{\infty} f[n-1] z^{-n}$$

$$m = n-1 \Rightarrow n = m+1$$

$$\sum_{m=-1}^{\infty} f[m] z^{-(m+1)}$$

$$= z \left[f[-1] z^{-1} + \sum_{n=0}^{\infty} f[n] z^{-n} \right]$$

$$y[n+1] - 2y[n] = 3^n ; y[0] = 2$$

$$\mathcal{Z}\{y[n+1]\} - 2\mathcal{Z}\{y[n]\} = \frac{z}{z-3} \because \mathcal{Z}(a^n) = \frac{z}{z-a}$$

$$z^2 \mathcal{Z}(y[n]) - z^2 y[0] - 2\mathcal{Z}(y[n]) = \frac{z}{z-3}$$

$$(z^2 - 2z) \mathcal{Z}(y[n])$$

$$\Rightarrow \{ z \mathcal{Z}(y[n]) - z(y[0]) \}$$

$$\Rightarrow (z-2) \mathcal{Z}(y[n]) - 2z(2) = \frac{z}{z-3}$$

$$\Rightarrow (z-2) \mathcal{Z}(y[n])$$

$$\Rightarrow \mathcal{Z} y[n]$$